



# **HANDBOOK OF COASTAL AND OCEAN ENGINEERING**

## **Expanded Edition**

Volume 1

edited by

**Young C Kim**

California State University, Los Angeles, USA

 **World Scientific**

NEW JERSEY • LONDON • SINGAPORE • BEIJING • SHANGHAI • HONG KONG • TAIPEI • CHENNAI • TOKYO

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# **HANDBOOK OF COASTAL AND OCEAN ENGINEERING**

## **Second Edition**

Volume 2

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*Published by*

World Scientific Publishing Co. Pte. Ltd.

5 Toh Tuck Link, Singapore 596224

USA office: 27 Warren Street, Suite 401-402, Hackensack, NJ 07601

UK office: 57 Shelton Street, Covent Garden, London WC2H 9HE

### **Library of Congress Cataloging-in-Publication Data**

Names: Kim, Young C., 1936– editor.

Title: Handbook of coastal and ocean engineering / edited by Young C. Kim,  
California State University, Los Angeles, USA.

Description: Expanded edition. | [Hackensack] New Jersey : World Scientific, 2017. |  
Includes bibliographical references.

Identifiers: LCCN 2017045719 | ISBN 9789813204010 (hardcover : set : alk. paper) |

ISBN 9789813204072 (hardcover : volume 1 : alk. paper) |

ISBN 9789813204089 (hardcover : volume 2 : alk. paper)

Subjects: LCSH: Coastal engineering--Handbooks, manuals, etc. |

Ocean engineering--Handbooks, manuals, etc.

Classification: LCC TC205 .H32 2017 | DDC 620/.4162--dc23

LC record available at <https://lcn.loc.gov/2017045719>

### **British Library Cataloguing-in-Publication Data**

A catalogue record for this book is available from the British Library.

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Typeset by Stallion Press

Email: [enquiries@stallionpress.com](mailto:enquiries@stallionpress.com)

Printed in Singapore

## Preface

Although coastal and ocean engineering is a very ancient field with the construction of Port A-ur near the mouth of the Nile in 3,000 BC, significant advances in this field have been made in the last several decades. The rise of interest in this field can be seen from the number of attendees by academics and practitioners in international conferences. The first International Conference on Coastal Engineering was held in Long Beach, California in 1950 with less than 100 people. When the same conference was held in San Diego, California in 2006, over 1,000 delegates attended. In the last several decades, the world has seen significant coastal and ocean engineering projects, one of which is the Delta Project in the Netherlands. This project was designed to shorten and strengthen the total length of coast and dykes washed by the sea by closing off the sea arms in the Delta region. Other noteworthy coastal engineering projects include the Kansai Airport Project in Japan and, in recent years, the construction of mobile barriers at inlets to regulate tides in the Venice Lagoon known as the Venice Project. Interest in coastal and ocean engineering has arisen in recent years as humankind experiences coastal disasters that derive from coastal storms, seismic and tsunami events, and the impact of climate change which has resulted in sea-level rise. The tsunami activity in Sumatra in December 2004 affected countries throughout the Indian Ocean and resulted in the loss of thousands of lives. Hurricane Katrina in New Orleans also claimed many lives with property damage exceeding \$63 billion. The 2011 Tohoku-Pacific ocean earthquake and tsunami in Japan caused extensive damage to properties and the loss of human lives. Global warming and sea-level rise will affect shoreline retreats, inundate low coastal areas, damage coastal structures, and accelerate beach erosion. The need for better understanding of our coastal and ocean environment has become of utmost importance in recent years.

With the successful sales of the *Handbook of Coastal and Ocean Engineering*, 1<sup>st</sup> edition in the last few years, the publishing company has decided to produce the expanded edition this year. For this edition, 41 chapters from the 1<sup>st</sup> edition will be reprinted and 18 new chapters will be added. From the 41 chapters in the 1<sup>st</sup> edition, some authors will be updating their manuscripts with new developments. There will be a total of 59 chapters in this expanded edition.

This handbook contains a comprehensive compilation of topics that are the forefront of many technical advances in coastal and ocean engineering. It represents the most comprehensive reference available on coastal and ocean engineering to date, and it also provides the most up-to-date technical advances and latest research findings on coastal and ocean engineering. More than 110 internationally recognized authorities in the field of coastal and ocean engineering contributed papers on their area of expertise to the handbook. These international luminaries are from highly respected universities and renowned research and consulting organizations from all over the world.

This handbook provides a comprehensive overview of shallow-water waves, water level fluctuations, coastal and offshore structures, wave-structure-soil interaction, ports and harbors, coastal sediment processes, environmental problems, sustainable coastal development, coastal hazards, ocean energy, physical modeling and numerical model validation, and coastal engineering practice and education. This book is an essential source of reference for professionals and researchers in the areas of coastal engineering, ocean engineering, oceanography, meteorology, and civil engineering, and as a text for graduate students in these fields. It will also be an excellent source book for educational and teaching purposes, and would be a great reference book for any technical library.

I would like to express my gratitude to the authors and co-authors who have contributed to this world-class handbook. It has been more than a year-long project, and their support and sacrifices have been deeply appreciated. I am truly grateful.

I would like to express my indebtedness to those who have guided me and supported me as a mentor and a colleague throughout my professional life. They are:

Professor Robert L. Wiegel, University of California, Berkeley

Professor Joe W. Johnson, University of California, Berkeley

Professor Robert G. Dean, University of Florida

Professor Fredric Raichlen, California Institute of Technology

Professor Raymond C. Binder, University of Southern California

Professor Shoshichiro Nagai, Osaka City University

Dr. Basil Wilson, Science Engineering Associates

Dr. Lars Skjelbreia, Science Engineering Associates

Dr. Bernard LeMehaute, University of Miami

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I also wish to express my indebtedness to those who nurtured me from my early teen years and changed the course of my life. They are:

Dr. Helen Miller Bailey, East Los Angeles College  
Mr. H. Karl Bouvier, Jet Propulsion Laboratory

I extend my gratitude to my wife, Janet, for her constant support, encouragement, patience, and understanding while I was undertaking the task, and to my daughter, Susan Calix, for proofreading some of the materials.

Finally, I wish to express my deep appreciation to Mr. Yeow-Hwa Quek and Ms. Chelsea Chin, Senior Editors of World Scientific Publishing Company who gave me invaluable support and encouragement from the inception of this handbook to its realization.

Young C. Kim  
Los Angeles, California  
August 2017

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**Young C. Kim, Ph.D., Dist.D.CE., F.ASCE**, is currently a Professor of Civil Engineering, Emeritus at California State University, Los Angeles. Other academic positions held include a Visiting Scholar of Coastal Engineering at the University of California, Berkeley (1971); a NATO Senior Fellow in Science at the Delft University of Technology (1975); a Visiting Scientist at the Osaka City University (1976); and a Visiting Professor, Polytech Nice-Sophia, Universite de Nice-Sophia Antipolis (2011-2013). For more than a decade, he served as Chair of the Department of Civil Engineering (1993-2005) and was Associate Dean of Engineering in 1978. For his dedicated teaching and outstanding professional activities, he was awarded the university-wide Outstanding Professor Award in 1994.

Dr. Kim was a consultant to the U.S. Naval Civil Engineering Laboratory in Port Hueneme and became a resident consultant to the Science Engineering Associates where he investigated wave forces on the Howard-Doris platform structure, now being placed in the Ninian Field, North Sea.

Dr. Kim is the past Chair of the Executive Committee of the Waterway, Port, Coastal and Ocean Division and served on the Executive Committee of the Technical Council on Research and the Civil Engineering Department Heads Council of the American Society of Civil Engineers (ASCE). He also served as Chair of the Nominating Committee of the International Association for Hydro-Environment Engineering and Research (IAHR). Since 1998, he served on the International Board of Directors of the Pacific Congress on Marine Science and Technology (PACON). He is the past President of PACON.

Dr. Kim has been involved in organizing 14 national and international conferences. He has authored seven books and published 55 technical papers in various engineering journals. Recently, he served as an editor for the *Handbook of Coastal and Ocean Engineering* (2010), *Coastal and Ocean Engineering Practice* (2012), and *Design of Coastal Structures and Sea Defenses* (2015), which were published by the World Scientific Publishing Company. In 2011, he was inducted as the Distinguished Diplomate of Coastal Engineering from the Academy of Coastal, Ocean, Port and Navigation Engineers (ACOPNE). In 2012, he was elected Fellow of the American Society of Civil Engineers.

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